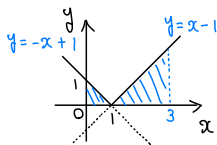


・絶対値を含む関数の定積分①

(例) 次の定積分を求めよ。

$$(1) \int_0^3 |x-1| dx \quad + \quad \left[\frac{1}{2}x^2 \times x \right]_0^3 \quad \left| \int_0^3 (x-1) dx \right|$$

$$|x-1| = \begin{cases} x-1 & (x \geq 1) \\ -(x-1) & (x < 1) \end{cases}$$

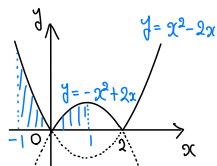


であるから

$$\begin{aligned} \int_0^3 |x-1| dx &= \int_0^1 -(x-1) dx + \int_1^3 (x-1) dx \\ &= \left[-\frac{1}{2}x^2 + x \right]_0^1 + \left[\frac{1}{2}x^2 - x \right]_1^3 \\ &= \frac{5}{2} // \end{aligned}$$

$$(2) \int_{-1}^1 |x^2 - 2x| dx$$

$$|x^2 - 2x| = \begin{cases} x^2 - 2x & (x \leq 0, 2 \leq x) \\ -(x^2 - 2x) & (0 < x < 2) \end{cases}$$



であるから

$$\begin{aligned} \int_{-1}^1 |x^2 - 2x| dx &= \int_{-1}^0 (x^2 - 2x) dx + \int_0^1 -(x^2 - 2x) dx \\ &= \left[\frac{1}{3}x^3 - x^2 \right]_{-1}^0 + \left[-\frac{1}{3}x^3 + x^2 \right]_0^1 \\ &= 2 // \end{aligned}$$